

MediBridge: Healthcare Access and Financial Assistance Platform

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1 Higher-order Goal

Healthcare accessibility remains a major challenge due to fragmented information, expensive treatment options, and difficulty in arranging financial support. This project aims to build MediBridge, a software platform that connects healthcare discovery and financial assistance into a unified ecosystem.

The engineering objective is to design a scalable web platform where patients can discover hospitals, manage medical information, understand treatment options, and access financial support workflows. The system focuses on improving decision-making and reducing delays during critical healthcare situations.

2 Time-to-value

The project follows an MVP-first development strategy:

- First iteration delivers patient registration, healthcare profiles, and hospital discovery.
- Subsequent iterations introduce medical record management and financial assistance workflows.
- Continuous testing and user feedback guide feature improvements.

3 Problem Statement

Patients often face difficulty finding affordable healthcare services. Information about hospitals, treatment costs, doctor availability, and financial assistance is usually distributed across multiple platforms.

Existing healthcare systems frequently focus on individual services rather than providing a

complete patient journey. Patients may struggle with maintaining medical records, comparing options, and arranging funds for treatment.

This project addresses the gap by developing a centralized healthcare assistance platform combining medical information access and financial support mechanisms.

4 Proposed Solution

4.1 Overview

MediBridge provides:

- Patient dashboard.
- Hospital and doctor search.
- Medical record storage.
- Treatment cost comparison.
- Loan and fundraising assistance workflows.
- Emergency health profile.

4.2 Core Workflow

1. Patient creates a secure profile.
2. User searches treatment options and hospitals.
3. Medical records are uploaded and managed.
4. Financial assistance requests can be initiated.
5. Emergency information can be accessed when required.

5 Solution Approach

5.1 System Architecture

The platform follows a three-tier architecture:

- **Frontend:** Responsive patient and administrator interfaces.
- **Backend:** REST APIs for authentication, workflows, and business logic.
- **Database:** Stores users, hospitals, records, and application history.

5.2 Security Approach

The system will implement authentication, authorization, secure document handling, and privacy-aware data storage.

5.3 CI/CD and Testing

Automated testing and deployment pipelines will enable reliable incremental development.

6 Evaluation Criterion

The system will be evaluated using:

- Patient workflow completion time.
- Recommendation usefulness.
- Document access reliability.
- User satisfaction scores.
- System availability.

7 Scalability

The architecture supports future growth through modular services, cloud deployment, database indexing, and API-based integrations with hospitals and financial organizations.

8 Engine Availability Heuristic

The project can be implemented using mature web frameworks, cloud databases, authentication systems, storage services, testing tools, and deployment platforms.

9 Project Scope and Deliverables

9.1 Initial Deliverable

- Authentication system.
- Patient dashboard.
- Hospital search.
- Medical profile management.

9.2 Subsequent Deliverables

- Financial assistance workflow.
- Recommendation engine.
- Notification system.
- Partner integrations.

10 Risks and Mitigations

Major risks include healthcare data privacy, large scope, and dependency on external healthcare information. These will be handled through limited MVP scope, secure storage practices, and modular development.

11 Summary

MediBridge proposes a scalable healthcare support platform that combines healthcare discovery and financial assistance into a single patient-focused solution.